

Shellfs

Read and write shell pipelines through file paths



Install & load

```
INSTALL shellfs FROM community;  
LOAD shellfs;
```

01 At a glance

Use a shell command as a DuckDB file source or sink. A trailing pipe reads command output; a leading pipe sends COPY output to a command. Useful for local tools that already produce or consume CSV and JSON.

Local command → Streaming pipe → DuckDB rows

02 Pipe paths + SQL integration

```
'command |'
```

Read the command's standard output as a file.

```
'| command'
```

Write bytes to the command's standard input.

```
read_csv(pipe_path, ...)
```

Parse streamed CSV with normal DuckDB reader options.

```
read_json(pipe_path, ...)
```

Parse JSON emitted by a command.

```
COPY (...) TO pipe_path
```

Stream query results into a command; specify the format.

```
SET ignore_sigpipe = true
```

Optional setting to ignore SIGPIPE; default is false.

03 Keep in mind

Pipe placement sets direction

Place `|` at the end to read, or at the beginning to write. Commands run through the host shell with the process's permissions. Keep paths trusted and never construct commands by concatenating untrusted input.

04 Query a command's CSV output

```
SELECT sum(n) AS total  
FROM read_csv(  
  'printf "n\n1\n2\n3\n" |',  
  header = true  
);  
-- 6
```

Runs local `printf`; the trailing pipe makes it a readable path.

05 Compress an export with gzip

```
COPY (SELECT range AS n FROM range(1, 4))  
TO '| gzip > /tmp/qf-shellfs.csv.gz'  
(FORMAT CSV, HEADER, COMPRESSION UNCOMPRESSED);
```

Requires `gzip` and writable `/tmp`. Disable COPY compression so only `gzip` compresses the stream.

06 Read the archive back

```
SELECT sum(n) AS total  
FROM read_csv('/tmp/qf-shellfs.csv.gz');  
-- 6
```

Run after the export above; DuckDB detects `gzip` from the filename.

07 In practice

Treat the source as a stream

Pipes do not behave like seekable local files. CSV and newline-delimited JSON are natural fits. Set explicit parsing options when needed, and ensure the command and any prerequisite tools exist on the host.

[Full docs & examples ↗](#) [Source ↗](#)